

ArbitrageAI Platform — Functional Requirements Document (FRD)

Classification: Internal — Engineering

Version: 1.0

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Prepared for: Technical Lead & Engineering Team

1. Authentication & User Management

ID	Function	Description	Acceptance Criteria
AUTH-01	Email registration	Register with email + password	Password min 8 chars; email verification within 30s
AUTH-02	Google SSO	One-tap sign in via Firebase Auth	Account created on first login; existing account detected on subsequent
AUTH-03	Apple SSO	Sign in with Apple (iOS required)	Compliant with Apple App Store requirement
AUTH-04	Session management	JWT token refresh	Tokens refresh silently; no force-logout during active session
AUTH-05	Multi-device	Web + mobile simultaneous login	Sessions independent
AUTH-06	Password reset	Email-based reset flow	Reset link expires 1 hour; single-use
PROF-01	Country selection	Home country set on onboarding	Defaults home currency based on country
PROF-02	Currency preference	ZAR, NGN, KES, USD display	Persisted in Firestore; applied to all P&L displays
PROF-03	Notification preferences	Per-channel enable/disable	Effective within 60s
PROF-04	Risk level	Conservative / Balanced / Aggressive	Maps to capital allocation weights
PROF-05	Capital allocation %	% of balance available to AI	Range 10%-100%; default 50%

2. Exchange Connectivity

ID	Function	Description	Acceptance Criteria
EXCH-01	Coinbase OAuth	OAuth 2.0 authorization code flow	Token in Secret Manager within 5s; balance shown within 3s
EXCH-02	Kraken OAuth	OAuth 2.0 flow	Same as EXCH-01
EXCH-03	Luno OAuth	OAuth 2.0 flow	Same as EXCH-01; Africa users prioritized
EXCH-04	Token refresh	Automatic OAuth token refresh	Refresh 24h before expiry; user notified on failure
EXCH-05	API key validation	Balance endpoint called on key paste	Connection confirmed within 2s
EXCH-06	Permission guidance	Show which permissions to enable/disable	Withdraw explicitly instructed OFF with screenshot
EXCH-07	Error handling	Human-readable error messages	No raw API error codes shown
EXCH-08	Disconnect detection	Monitor WebSocket health	Reconnect within 10s; user notified after 30s failure
EXCH-09	Balance refresh	Real-time balance display	Updated every 60s; stale indicator if >90s
EXCH-10	Disconnect exchange	Remove exchange connection	API key/token deleted from Secret Manager within 5s
EXCH-11	Reconnect prompt	Alert on expired connection	WhatsApp/Telegram alert + reconnect button
EXCH-12	Key rotation prompt	90-day rotation reminder	Wizard guides new key generation

3. Strategy Engine

ID	Function	Description	Acceptance Criteria
STRAT-01	AI engine toggle	Activate/pause all strategies	State changes within 5s; all worker pods start/pause
STRAT-02	Per-exchange toggle	Enable/disable individual exchanges	Takes effect within 30s; no open positions left
STRAT-03	Risk level application	Risk slider adjusts capital allocation	Applied within 60s of change
STRAT-04	Strategy availability by tier	Tier-gated strategy access	Locked strategies show "Upgrade to unlock"
OPP-01	Funding rate polling	Poll Binance, Bybit, OKX funding rates	Every 8 hours; annualized rate calculated

ID	Function	Description	Acceptance Criteria
OPP-02	Spatial spread detection	Real-time WebSocket price comparison	Spread detected within 50ms
OPP-03	Africa P2P monitoring	Monitor VALR, Luno, Quidax vs. Binance	Every 5 minutes; ZAR and NGN tracked separately
OPP-04	Naira devaluation detector	Monitor CBN rate vs. P2P rate gap	Alert when gap > 5%
EXEC-01	Profitability pre-filter	Net profit calc before every trade	Trade rejected if < 0.15% after all fees
EXEC-02	Slippage guard	Check slippage before order submission	Reject if actual slippage > tier threshold
EXEC-03	Order placement	Submit via Exchange Connector gRPC	Within 100ms of green-light
EXEC-04	Partial fill handling	Handle < 80% fill within 500ms	Cancel remainder; close hedge immediately
EXEC-05	Hedge closure	Close opposing leg on partial fill	No unhedged legs ever left open
EXEC-06	Execution quality recording	Record expected vs. actual fill	actual_vs_expected_ratio written to Cloud Spanner
RISK-01	Daily loss limit	Auto-pause if daily P&L < threshold	Pause within 5s; WhatsApp/SMS alert sent
RISK-02	Capital per trade limit	Hard limit enforced in order router	Cannot be overridden by AI model
RISK-03	Exchange concentration	Max % on single exchange	Checked before every order
RISK-04	KILL_SWITCH	Platform-wide strategy pause	All Pub/Sub subscriptions paused within 10s
RISK-05	Strategy deprioritization	7-day underperformance → weight = 0%	Calculated daily by BigQuery job

4. FX Rate Service

ID	Function	Description	Acceptance Criteria
FX-01	Dashboard FX refresh	Update local currency P&L	Every 60s from Open Exchange Rates API
FX-02	Execution FX capture	Store FX rate at trade close	Written atomically with trade record in Cloud Spanner
FX-03	Nigeria multi-rate display	CBN official, Wise parallel, P2P rate	All three shown; user selects preferred; default P2P
FX-04	FX fallback	Switch to Wise API if primary unavailable	Automatic failover within 5s

5. Backtesting

ID	Function	Description	Acceptance Criteria
BT-01	Historical data ingestion	OHLCV + funding rates from Tardis.dev	Loaded into BigQuery; partitioned by exchange, asset, date
BT-02	Africa P2P data collection	Scrape P2P premium every 15 minutes	Cloud Function from Day 1; data stored in BigQuery
BT-03	Backtest simulation	Strategy vs. historical data	Results in backtest_results table
BT-04	Performance gate check	Auto-evaluate vs. minimum thresholds	Sharpe > 1.5, drawdown < 10%, win rate > 65%, 6+ months data
BT-05	Gate enforcement	Block live deployment of failing strategies	Strategy marked live_approved: false; deployment halted
BT-06	Paper trading mode	Live data, no real orders	Simulated P&L in separate BigQuery table

6. Notifications

ID	Function	Description	Acceptance Criteria
NOTIF-01	WhatsApp trade alert	Trade executed alert via WhatsApp	Within 5s of trade close; NGN/ZAR/KES amount shown
NOTIF-02	Telegram trade alert	Trade alert via Telegram Bot	Within 5s; all strategy details included
NOTIF-03	SMS critical alert	SMS for kill switch + loss limit events	Via Twilio within 10s
NOTIF-04	Daily P&L summary	Morning summary at 8:00 AM local time	Skipped if P&L = \$0
NOTIF-05	Quiet market message	Proactive message if no trades for 48h	Explains market conditions
NOTIF-06	Telegram commands	/pnl, /status, /pause, /resume, /balance	Response within 3s
NOTIF-07	Channel routing	Country-based channel selection	NG/KE/GH → WhatsApp; ZA → WhatsApp then Telegram

7. Dashboard & Reporting

ID	Function	Description	Acceptance Criteria
DASH-01	Real-time portfolio value	Total balance in local currency + USD	Updated every 60s
DASH-02	Today's P&L	Dollar amount + percentage	Updated within 3s of trade close

ID	Function	Description	Acceptance Criteria
DASH-03	Active strategy indicators	Live funding rate %, spread %, P2P %	Updated every 60s
DASH-04	P&L history chart	Daily/weekly/monthly Recharts line chart	Timezone-adjusted to user's local time
DASH-05	Trade log	Per-trade detail; paginated (20/page)	Exportable to CSV
DASH-06	SARS tax export	SARS-compliant CSV with ZAR cost base + proceeds	Uses fx_rate_at_execution; South Africa users only
DASH-07	Per-strategy P&L	Breakdown by strategy type	MTD and all-time views

8. Subscription & Billing

ID	Function	Description	Acceptance Criteria
BILL-01	Free trial	14-day Pro trial, no payment	Starts on account creation; ends exactly Day 15
BILL-02	Trial end flow	Engine paused at Day 15 without payment	Dashboard read-only; upgrade prompt with trial P&L
BILL-03	Stripe checkout	Card payment	Subscription active within 60s
BILL-04	Flutterwave checkout	Africa bank transfer + mobile money	Confirmed via webhook; subscription activated
BILL-05	M-Pesa checkout	Kenya mobile money	STK push initiated; activated on confirmation
BILL-06	Tier upgrade	Starter → Pro	Workers upgraded within 60s; new strategies available
BILL-07	Tier downgrade	Pro → Starter	Pro strategies paused at end of billing period
BILL-08	Cancellation retention	P&L-based retention message	Positive P&L → show earnings; zero/negative → offer free month
BILL-09	Subscription webhook	Stripe/Flutterwave events update Firestore	subscription_status updated within 30s

9. Admin & Operations

ID	Function	Description	Acceptance Criteria
ADMIN-01	Global kill switch	Pause all strategies platform-wide	All workers paused within 10s
ADMIN-02	Per-user kill switch	Pause specific user's strategies	User notified; paused within 5s

ID	Function	Description	Acceptance Criteria
ADMIN-03	Strategy approval gate	Manual approval after backtest pass	No automated live deployment without admin approval
ADMIN-04	Exchange blacklist	Blacklist exchange platform-wide	All users' strategies on that exchange paused within 60s
ADMIN-05	User tier override	Manually set user tier	Effective immediately
ADMIN-06	Audit log	All admin actions logged	Immutable; viewable in Cloud Console

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